THE SOUTH AFRICAN MATHEMATICS OLYMPIAD

Senior Second Round 2025

Solutions

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1. **Answer 025**

   $(2025)^2 - 2(2025)(2020) + (2020)^2 = (2025 - 2020)^2 = 25$.

2. **Answer 005**

   First, $2^x = 8 = 2^3$, so $x = 3$. Then $8^y = 16$, giving $2^{3y} = 2^4$, so $3y = 4$ and $y = \frac{4}{3}$. Similarly, $16^z = 32$ gives $z = \frac{5}{4}$, so $xyz = 3 \times \frac{4}{3} \times \frac{5}{4} = 5$.

3. **Answer 002**

   If $a =$ apples and $b =$ bananas, then $5a + 5b = 2b + 11a$, so $3b = 6a$ and $b = 2a$, meaning that one banana is worth two apples.

4. **Answer 022**

   The bottom layer needs one more cube, the second layer needs two cubes, the third layer seven cubes and the top layer 12 cubes, making a total of 22 extra cubes.

5. **Answer 003**

   If the bucket weighs b kg and the water w kg, then $b + w = 21$ and $b + \frac{1}{2}w = 12$. Subtracting gives $\frac{1}{2}w = 9$, so the water weighs 18 kg and the empty bucket 3 kg.

6. **Answer 016**

   If the digits of the original number are A and B respectively, then the units digit of $A \times B$ must be 6, to match the units digit of 976. Since $A > B$, it follows that A and B must be 6 and 1 or 3 and 2, having only $6 + 1 = 7$. To confirm, check that $61 \times 16 = 976$.

7. **Answer 004**

   The diameter of the circles is the height of the rectangle, which is 5. The distance between their centres is equal to the width of the rectangle minus two radii (or one diameter), that is, $9 - 5$, which is 4.

8. **Answer 006**

If the distance one way is d km, then $\frac{d}{4} + \frac{d}{3} = 3\frac{1}{2}$, so $\frac{7}{12}d = \frac{7}{2}$, giving $d = 6$.

9. **Answer 012**

In one hour, the old tractor ploughs $\frac{1}{10}$ hectare and the new one ploughs $\frac{1}{5}$ hectare. Together, they plough $\frac{1}{10} + \frac{1}{5} = \frac{3}{10}$ hectare per hour, so in two hours they plough $\frac{6}{10} = \frac{3}{5}$ hectares, leaving $2\frac{2}{5} = \frac{12}{5}$ hectares to be ploughed. At $\frac{1}{5}$ hectare per hour, the new tractor will take 12 more hours to finish the ploughing.

10. **Answer 096**

If students A and B are treated as an item (in either order), and similarly with C and D, then together with E and F there are four items, which can be arranged in $4\times3\times2\times1 = 24$ different ways. With each of these arrangements the items AB and CD can each be ordered in two different ways, giving a total of $24 \times 2 \times 2 = 96$ ways for the six students to sit together.

11. **Answer 297**

The area of a trapezium is average width times height, so $300 = \frac{1}{2}(a + b)h$. Increasing the height by 10% gives a new height of $1.1h$, and decreasing the width by 10% gives a new average width of $0.9 \times \frac{1}{2}(a + b)$. Thus the area of the new trapezium is $(1.1)(0.9)(\frac{1}{2})(a + b)h = 0.99 \times 300 = 297$.

12. **Answer 002**

From the expression one can see that $x = \sqrt{6 - x}$, which is positive. Squaring leads to $x^2 = 6 - x$ or $(x - 2)(x + 3) = 0$, for which the only positive solution is $x = 2$.

13. **Answer 028**

There are eight different orderings of the children, from BBB to GGG, which each have probability $\frac{1}{8}$ since boys and girls are equally likely. Among these there are three with two girls (BGG, GBG, GGB) and one with three girls (GGG), so the probability of at least two girls is $\frac{4}{8} = \frac{1}{2}$. Thus $\frac{x}{56} = \frac{1}{2}$, giving $x = 28$.

14. **Answer 020**

Suppose the numbers chosen are a, b, c, d and let $S = a + b + c + d$. We are given that $a + \frac{2}{3}(b + c + d) = 42$, which can be written as $\frac{1}{3}a + \frac{2}{3}S = 42$. Adding this equation to the three similar ones gives $\frac{1}{3}S + \frac{8}{3}S = 42 + 54 + 66 + 78$, which simplifies to $3S = 240$. Thus $S = 80$ and the mean $\frac{1}{4}(a + b + c + d) = \frac{1}{4}S = 20$.

15. **Answer 144**

The region has the same area as rectangle $AFDC$, since the upper portion has three segments removed from square $AFEB$ and the lower portion has three identical segments added to square $BEDC$. The area of the rectangle is $AC \times CD = 2BC \times CD$. Since $BC = CD$, the area can be written as $BC^2 + CD^2$, which equals BD^2 by Pythagoras' theorem, and $BD^2 = (6 + 6)^2 = 144$.

16. **Answer 191**

The number of unordered selections of two songs from a list of 20 is $\frac{20 \times 19}{2} = 190$, because there are 20 choices for one song and 19 choices for the other, divided by 2 because the order of the songs does not matter. To guarantee that at least one selection is repeated, the number of contestants must be greater than the number of selections, so the minimum number of contestants is 191.

17. **Answer 720**

If $S = a + b + c + d + e + f$, then the first equation can be written as $S - f = 20$, and similarly with the others. Adding all six equations then gives $6S - S = 20 + 19 + 18 + 17 + 16 + 15$, or $5S = 105$, giving $S = 21$. Thus $f = 1$, $e = 2$, and so on, and the product $a \times b \times c \times d \times e \times f = 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 720$.

18. **Answer 006**

For the fewest even numbers there must be the most odd numbers, and they must be as large as possible. The sum of all 50 odd numbers from 1 to 99 is $50^2 = 2500$, which is 400 short of 2900. Thus more than four even numbers are required, since they do not exceed 100. Dropping the six smallest odd numbers, from 1 to 11, reduces the total to 2464, which can be brought up to 2900 by replacing the odd numbers from 1 to 11 with any six even numbers adding up to 436.

19. **Answer 592**

The product $15 \times n$, using only 8's and 0's, is an even number divisible by 5 and 3, so its last digit must be 0 and the sum of its digits must be divisible by 3. The smallest possibility is $8880 = 15 \times 592$.

20. **Answer 099**

Each internal angle of a regular pentagon is $180° - \frac{1}{5} \times 360° = 108°$, and for a regular octagon each is $180° - \frac{1}{8} \times 360° = 135°$. Triangle AEB is isosceles with $E\widehat{A}B = 108°$, so $E\widehat{B}A = \frac{1}{2}(180° - 108°) = 36°$. Then $B\widehat{A}T = E\widehat{B}A = 36°$ by alternate angles, since $AT \parallel EB \parallel DW$. Finally, $Z\widehat{A}B = Z\widehat{A}T - B\widehat{A}T = 135° - 36° = 99°$.

21. **Answer 096**

The first cube has one hole on each face, the second cube has four holes, and in general the n-th cube has n^2 holes on each face, each hole being the size of $3^2 = 9$ small cubes. Similarly, the first big cube has five small cubes along each edge, the next one nine, and so on, so there are $4n + 1$ small cubes along each edge of the n-th one. The number of small cubes on one face of the n-th big cube is therefore $(4n+1)^2 - 9n^2 = (7n+1)(n+1)$. The total number of small cubes on all six faces is then approximately $6 \times 7n^2 = 42n^2$. (Small cubes on edges and vertices have been counted more than once, but an approximation is adequate, since an integer solution is required.) The equation $42n^2 = 674$ simplifies to $n^2 \approx 16$, so $n = 4$ and the total number of holes on the six faces is $6n^2 = 6 \times 16 = 96$.

Alternative solution:

The side length of element n is $4n + 1$ small cubes, which gives a total of $(4n+1)^3 = 64n^3 + 48n^2 + 12n + 1$ small cubes in a solid large cube of the same size. The inner large cube, leaving a width of 1 small cube on all the faces of the large cube, will have a side length of $4n - 1$ small cubes, yielding $(4n-1)^3 = 64n^3 - 48n^2 + 12n - 1$ small cubes in the inner large cube. The holes in the large cube's faces takes up $6 \times 9 \times n^2 = 54n^2$ small cubes. This leaves $64n^3 + 48n^2 + 12n + 1 - 64n^3 + 48n^2 - 12n + 1 - 54n^2 = 42n^2 + 2$ small cubes in the structure of the n-th hollow cube. Thus, if $42n^2 + 2 = 674$, then $42n^2 = 672$ giving $n^2 = 16$ and therefore $n = 4$. The total number of holes in all the faces of the fourth element is $6 \times 4^2 = 6 \times 16 = 96$.

22. **Answer 012**

The longest chord through P is the diameter of the circle, of length 30. The shortest chord is the one perpendicular to the diameter through P, and P is its midpoint. By Pythagoras' theorem, half the length of this (shortest) chord is $\sqrt{15^2 - 9^2} = 3\sqrt{5^2 - 3^2} = 3 \times 4 = 12$. Thus the shortest chord has length 24, and the number of chords from the shortest to the longest chord is the number of integers n satisfying $24 \leq n \leq 30$, which is 7. However, all of these chords, except the diameter and the shortest chord, can be mirrored with respect to the diameter. This yields a total of $2 \times (7 - 2) + 2 = 12$ different chords.

23. **Answer 028**

If both sides of the equation are increased by 1, then it can be rewritten as $(a + 1)(b + 1)(c + 1) = 1001 = 7 \times 11 \times 13$. Thus $a + b + c = 6 + 10 + 12 = 28$.

24. **Answer 998**

Let's start with values from $n = 1000$ and search for a pattern when $n < 1000$.

$$\begin{aligned}
f(1000) &= 997 \\
f(999) = f(f(1004)) &= f(1001) = 998 \\
f(998) = f(f(1003)) &= f(1000) = 997 \\
f(997) = f(f(1002)) = f(999) &= f(f(1004)) = f(1001) = 998 \\
f(996) = f(f(1001)) = f(998) &= f(f(1003)) = f(1000) = 997 \\
f(995) = f(f(1000)) = f(997) &= f(f(1002)) = f(999) = f(f(1004)) = f(1001) = 998
\end{aligned}$$

From the above, the following can be deduced (using induction):

$$f(n) = \begin{cases} 997 & \text{if } n \text{ is even and } n < 1000 \\ 998 & \text{if } n \text{ is odd and } n < 1000 \end{cases}$$

Now, for $n < 995$,

$$f(n) = f(f(n + 5)) = \begin{cases} f(997) = 998 & \text{if } n + 5 \text{ is even} \\ f(998) = 997 & \text{if } n + 5 \text{ is odd} \end{cases}$$

Note that n is even when $n + 5$ is odd, and n is odd when $n + 5$ is even.

$$\therefore f(25) = 998$$

25. **Answer 040**

Make use of Cartesian coordinates, with $D = (0, 0)$. Since $EF = CE$ and $C\widehat{E}F = 90°$, we have $F = (4, 12)$ and $K = (2, 8)$. $EK = FK$ giving $AK = 2$, thus $CK = \sqrt{6^2 + 8^2} = 10$ and it follows that $L = (2+8, 8+6) = (10, 14)$. Now we can simply calculate the difference in horizontal and vertical coordinates between F and L and use Pythagoras' theorem. Hence $FL^2 = (10 - 4)^2 + (14 - 12)^2 = 6^2 + 2^2 = 40$.

Alternative solution (using trigonometry):

By Pythagoras' theorem $CE^2 = CD^2 + CE^2 = 8^2 + 4^2 = 80$, so $CD = \sqrt{80} = 4\sqrt{5}$. Thus each square is obtained from the previous one by rotating through an angle α, where $\tan\alpha = \frac{4}{8} = \frac{1}{2}$, and scaling by a factor $\frac{4\sqrt{5}}{8} = \frac{\sqrt{5}}{2}$. By similar triangles it is easy to see that $AK = 2$ and $KF = 2\sqrt{5}$, and scaling up by $\frac{\sqrt{5}}{2}$ gives $FP = \sqrt{5}$ and $PL = 5$. Angle $F\widehat{P}L = A\widehat{K}F = 180° - (90° - \alpha) = 90° + \alpha$, so $\cos F\widehat{P}L = -\sin\alpha = -\frac{1}{\sqrt{5}}$. Finally, by the Cosine Rule,

$FL^2 = FP^2 + PL^2 - 2 \cdot FP \cdot PL \cdot \cos F\widehat{P}L = 5 + 25 - 2 \cdot 5 \cdot \sqrt{5} \cdot (\frac{-1}{\sqrt{5}}) = 30 + 10 = 40.$

25. **Antwoord 040**

Maak gebruik van Cartesiese koördinate, met $D = (0, 0)$. Synde $EF = CE$ en $C\widehat{E}F = 90°$, het ons $F = (4, 12)$ en $K = (2, 8)$. $EK = FK$ gee $AK = 2$, dus is $CK = \sqrt{6^2 + 8^2} = 10$ en dit volg dat $L = (2+8, 8+6) = (10, 14)$. Nou kan ons eenvoudig die verskil tussen die horisontale en vertikale koördinate van F en L bereken en dan Pythagoras se stelling gebruik. Gevolglik is $FL^2 = (10-4)^2 + (14-12)^2 = 6^2 + 2^2 = 40$.

Alternatiewe oplossing (gebruik trigonometrie):

Volgens Pythagoras se stelling is $CE^2 = CD^2 + CE^2 = 8^2 + 4^2 = 80$, sodat $CD = \sqrt{80} = 4\sqrt{5}$. Elke vierkant word verkry deur die vorige een deur 'n hoek van α te roteer, waar $\tan\alpha = \frac{4}{8} = \frac{1}{2}$, en te skaleer met 'n faktor van $\frac{4\sqrt{5}}{8} = \frac{\sqrt{5}}{2}$. Deur gelyksoortige driehoeke te gebruik is dit maklik om te sien dat $AK = 2$ en $KF = 2\sqrt{5}$, en 'n skalering van $\frac{\sqrt{5}}{2}$ gee $FP = \sqrt{5}$ en $PL = 5$. Hoek $F\widehat{P}L = A\widehat{K}F = 180° - (90° - \alpha) = 90° + \alpha$, so $\cos F\widehat{P}L = -\sin\alpha = -\frac{1}{\sqrt{5}}$. Deur die kosinusreël te gebruik, volg dit dat $FL^2 = FP^2 + PL^2 - 2 \cdot FP \cdot PL \cdot \cos F\widehat{P}L = 5 + 25 - 2 \cdot 5 \cdot \sqrt{5} \cdot (\frac{-1}{\sqrt{5}}) = 30 + 10 = 40$.

22. **Antwoord 012**

Die langste koord deur P is die middellyn van die sirkel, met lengte 30. Die kortste koord is die een loodreg op die middellyn deur P, en P is sy middelpunt. Volgens Pythagoras se stelling, is die helfte van die lengte van hierdie (kortste) koord $\sqrt{15^2 - 9^2} = 3\sqrt{5^2 - 3^2} = 3 \times 4 = 12$. Die kortste koord se lengte is dus 24, en die aantal koorde van die kortste tot die langste koord is die aantal heelgetalle n sodat $24 \leq n \leq 30$, wat 7 is. Al hierdie koorde, behalwe die middellyn self en die kortste koord, kan egter relatief tot die middellyn gereflekteer word. Dit lewer 'n totaal van $2 \times (7 - 2) + 2 = 12$ verskillende koorde.

23. **Antwoord 028**

As albei kante van die vergelyking met 1 vermeerder word, kan dit herskryf word as $(a + 1)(b + 1)(c + 1) = 1001 = 7 \times 11 \times 13$. Dus is $a + b + c = 6 + 10 + 12 = 28$.

24. **Antwoord 998**

Begin met waardes vanaf $n = 1000$ om 'n patroon te vind wanneer $n < 1000$.

$$\begin{aligned}
f(1000) &= 997 \\
f(999) = f(f(1004)) &= f(1001) = 998 \\
f(998) = f(f(1003)) &= f(1000) = 997 \\
f(997) = f(f(1002)) = f(999) &= f(f(1004)) = f(1001) = 998 \\
f(996) = f(f(1001)) = f(998) &= f(f(1003)) = f(1000) = 997 \\
f(995) = f(f(1000)) = f(997) &= f(f(1002)) = f(999) = f(f(1004)) = f(1001) = 998
\end{aligned}$$

Uit bostaande kan die volgende afleiding gemaak word (deur induksie te gebruik):

$$f(n) = \begin{cases} 997 & \text{as } n \text{ ewe is en } n < 1000 \\ 998 & \text{as } n \text{ onewe is en } n < 1000 \end{cases}$$

Nou, vir $n < 995$,

$$f(n) = f(f(n + 5)) = \begin{cases} f(997) = 998 & \text{as } n + 5 \text{ ewe is} \\ f(998) = 997 & \text{as } n + 5 \text{ onewe is} \end{cases}$$

Let daarop dat n ewe is wanneer $n + 5$ onewe is, en n is onewe wanneer $n + 5$ ewe is.

$$\therefore f(25) = 998$$

19. **Antwoord 592**

Die produk $15 \times n$, wat slegs uit 8's en 0'e bestaan, is 'n ewe getal wat deelbaar is deur beide 5 en 3, so die laaste syfer moet 0 wees en die som van die syfers deelbaar deur 3. Die kleinste moontlikheid hiervoor is $8880 = 15 \times 592$.

20. **Antwoord 099**

Elke binnehoek van 'n reëlmatige vyfhoek is $180° - \frac{1}{5} \times 360° = 108°$, en vir 'n reëlmatige agthoek is dit $180° - \frac{1}{8} \times 360° = 135°$. Driehoek AEB is gelykbenig met $E\widehat{A}B = 108°$, sodat $E\widehat{B}A = \frac{1}{2}(180° - 108°) = 36°$. Dan is $B\widehat{A}T = E\widehat{B}A = 36°$ omdat dit verwisselende hoeke is, want $AT \parallel EB \parallel DW$. Dit volg dat $Z\widehat{A}B = Z\widehat{A}T - B\widehat{A}T = 135° - 36° = 99°$.

21. **Antwoord 096**

Die eerste kubus het een gat op elke vlak, die tweede kubus vier gate en in die algemeen het die n-de kubus n^2 gate op elke vlak, waar elke gat die grootte van $3^2 = 9$ klein kubusse beslaan. Soortgelyk het die eerste groot kubus vyf klein kubusse op elke rand, die volgende een nege, ensovoorts, so daar is $4n + 1$ klein kubusse op elke rand van die n-de element. Die aantal klein kubusse op een vlak van die n-de groot kubus is vervolgens $(4n + 1)^2 - 9n^2 = (7n + 1)(n + 1)$. Die totale aantal klein kubusse op al ses vlakke saam is dan by benadering $6 \times 7n^2 = 42n^2$. (Van die klein kubusse op die rande en hoekpunte word meer as een keer getel, maar 'n benadering is voldoende aangesien 'n heelgetal as oplossing benodig word.) Die vergelyking $42n^2 = 674$ vereenvoudig na $n^2 \approx 16$, sodat $n = 4$ en die totale aantal gate op die ses vlakke saam gelyk is aan $6n^2 = 6 \times 16 = 96$.

Alternatiewe oplossing:

Die sylengte van element n is $4n + 1$ klein kubusse, wat 'n totaal van $(4n + 1)^3 = 64n^3 + 48n^2 + 12n + 1$ klein kubusse gee waaruit 'n soliede groot kubus van dieselfde grootte sal bestaan. Die binneste groot kubus, wat 'n dikte van 1 klein kubus op al die vlakke van die groot kubus oorlaat, sal 'n sylengte van $4n - 1$ klein kubusse hê, wat gee dat die binneste groot kubus uit $(4n - 1)^3 = 64n^3 - 48n^2 + 12n - 1$ klein kubusse sal bestaan. Die gate in die groot kubus se vlakke neem $6 \times 9 \times n^2 = 54n^2$ klein kubusse in beslag. Dit los dan $64n^3 + 48n^2 + 12n + 1 - 64n^3 + 48n^2 - 12n + 1 - 54n^2 = 42n^2 + 2$ klein kubusse in die struktuur van die n-de hol kubus oor. Dus, as $42n^2 + 2 = 674$, dan is $42n^2 = 672$ wat $n^2 = 16$ gee en daarvolgens is $n = 4$. Die totale aantal gate in al die vlakke van die vierde element saam, is $6 \times 4^2 = 6 \times 16 = 96$.

14. **Antwoord 020**

Gestel die getalle wat gekies is, is a, b, c, d en laat $S = a + b + c + d$. Dit word gegee dat $a + \frac{2}{3}(b + c + d) = 42$, wat ook as $\frac{1}{3}a + \frac{2}{3}S = 42$ geskryf kan word. Tel hierdie vergelyking by die ander drie soortgelyke vergelykings en verkry $\frac{1}{3}S + \frac{8}{3}S = 42 + 54 + 66 + 78$, wat vereenvoudig na $3S = 240$. Dus is $S = 80$ en die gemiddeld $\frac{1}{4}(a + b + c + d) = \frac{1}{4}S = 20$.

15. **Antwoord 144**

Die vaas het dieselfde oppervlakte as reghoek $AFDC$, aangesien daar drie segmente by die boonste vierkant $AFEB$ verwyder is en drie identiese segmente by die onderste vierkant $BEDC$ gevoeg is. Die oppervlakte van die reghoek is $AC \times CD = 2BC \times CD$. Omdat $BC = CD$, kan die oppervlakte herskryf word as $BC^2 + CD^2$, wat volgens Pythagoras se stelling gelyk is aan BD^2, en $BD^2 = (6 + 6)^2 = 144$.

16. **Antwoord 191**

Die aantal ongeordende keuses van twee liedjies uit 'n lys van 20 is $\frac{20 \times 19}{2} = 190$, want daar is 20 opsies vir die een liedjie en 19 opsies vir die ander een, en dit word deur 2 gedeel omdat die volgorde van die liedjies nie saak maak nie. Om verseker te wees dat minstens een seleksie herhaal word, moet die aantal deelnemers groter wees as die aantal moontlike keuses, so die kleinste getal deelnemers is 191.

17. **Antwoord 720**

Gestel $S = a+b+c+d+e+f$, dan kan die eerste vergelyking geskryf word as $S-f = 20$, en soortgelyk vir die ander vergelykings. Wanneer al ses vergelykings bymekaar getel word, gee dit $6S - S = 20 + 19 + 18 + 17 + 16 + 15$, oftewel $5S = 105$, wat $S = 21$ gee. Dus is $f = 1$, $e = 2$, ensovoorts, en die produk $a \times b \times c \times d \times e \times f = 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 720$.

18. **Antwoord 006**

Om die kleinste aantal ewe getalle te gebruik moet die grootste aantal onewe getalle, elkeen so groot as moontlik, gebruik word. Die som van al 50 onewe getalle van 1 tot 99 is $50^2 = 2500$, wat 400 minder as 2900 is. Dus word meer as vier ewe getalle benodig omdat nie een van hulle groter as 100 mag wees nie. Verwyder die ses kleinste ewe getalle van 1 tot 11, dan verminder die totaal na 2464, wat tot by 2900 verhoog kan word deur die onewe getalle van 1 tot 11 met enige ses ewe getalle wat optel na 436 te vervang.

8. **Antwoord 006**

   As die afstand in een rigting d km is, dan is $\frac{d}{4} + \frac{d}{3} = 3\frac{1}{2}$, dus $\frac{7}{12}d = \frac{7}{2}$, wat $d = 6$ gee.

9. **Antwoord 012**

   In een uur ploeg die ou trekker $\frac{1}{10}$ hektaar en die nuwe een ploeg $\frac{1}{5}$ hektaar. Gesamentlik ploeg hulle $\frac{1}{10} + \frac{1}{5} = \frac{3}{10}$ hektaar per uur, so in twee ure dus $\frac{6}{10} = \frac{3}{5}$ hektaar, wat $2\frac{2}{5} = \frac{12}{5}$ hektaar oorlaat wat nog geploeg moet word. Teen $\frac{1}{5}$ hektaar per uur sal dit die nuwe trekker nog 12 uur neem om klaar te ploeg.

10. **Antwoord 096**

    As studente A en B as een item (in enige volgorde) hanteer word en dieselfde met C en D, dan is daar vier items wanneer E en F bygevoeg word, en dit kan op $4 \times 3 \times 2 \times 1 = 24$ verskillende maniere gerangskik word. Vir elk van hierdie rangskikkings kan die items AB en CD elkeen op twee verskillende maniere gerangskik word, wat 'n totaal van $24 \times 2 \times 2 = 96$ maniere gee hoe die ses studente by mekaar kan sit.

11. **Antwoord 297**

    Die oppervlakte van 'n trapesium is die gemiddelde breedte gemaal met die hoogte, so $300 = \frac{1}{2}(a + b)h$. Deur die hoogte met 10% te vergroot gee 'n nuwe hoogte van $1.1h$, en wanneer die breedte met 10% verklein word gee dit 'n nuwe gemiddelde breedte van $0.9 \times \frac{1}{2}(a + b)$. Die oppervlakte van die nuwe trapesium is vervolgens $(1.1)(0.9)(\frac{1}{2})(a + b)h = 0.99 \times 300 = 297$.

12. **Antwoord 002**

    Uit die uitdrukking kan gesien word dat $x = \sqrt{6 - x}$, wat positief is. Kwadrering lewer $x^2 = 6 - x$ oftewel $(x - 2)(x + 3) = 0$, waarvoor die enigste positiewe oplossing $x = 2$ is.

13. **Antwoord 028**

    Daar bestaan agt verskillende volgordes vir die kinders, van BBB tot GGG (B stel 'n seun voor en G 'n dogter), wat elkeen 'n waarskynlikheid van $\frac{1}{8}$ het, aangesien die kanse vir 'n seun dieselfde is as die kanse vir 'n dogter. Uit hierdie moontlikhede is daar drie met twee dogters (BGG, GBG, GGB) en een met drie dogters (GGG), en dan is die waarskynlikheid vir minstens twee dogters $\frac{4}{8} = \frac{1}{2}$. Dus is $\frac{x}{56} = \frac{1}{2}$, wat $x = 28$ gee.

DIE SUID-AFRIKAANSE WISKUNDE OLIMPIADE

Senior Tweede Rondte 2025

Oplossings

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1. **Antwoord 025**

   $(2025)^2 - 2(2025)(2020) + (2020)^2 = (2025 - 2020)^2 = 25$.

2. **Antwoord 005**

   Eerstens is $2^x = 8 = 2^3$, dus $x = 3$. Dan is $8^y = 16$, wat $2^{3y} = 2^4$ gee, en dus $3y = 4$ en $y = \frac{4}{3}$. Soortgelyk, $16^z = 32$ gee $z = \frac{5}{4}$, sodat $xyz = 3 \times \frac{4}{3} \times \frac{5}{4} = 5$.

3. **Antwoord 002**

   As $a =$ appels en $b =$ piesangs, dan is $5a + 5b = 2b + 11a$, sodat $3b = 6a$ en $b = 2a$, wat beteken een piesang is twee appels werd.

4. **Antwoord 022**

   Die onderste laag kort nog een kubus, die tweede laag kort twee, die derde laag sewe kubusse en die boonste laag 12 kubusse, wat 'n totaal van 22 ekstra kubusse gee.

5. **Antwoord 003**

   As die emmer se gewig b kg is en die water se gewig w kg, dan is $b + w = 21$ en $b + \frac{1}{2}w = 12$. Deur aftrekking verkry ons $\frac{1}{2}w = 9$, so die water weeg 18 kg en die leë emmer 3 kg.

6. **Antwoord 016**

   As die syfers van die oorspronklike getal onderskeidelik A en B is, dan moet die ene-syfer van $A \times B$ gelyk aan 6 wees om met die ene-syfer van 976 ooreen te stem. Synde $A > B$, volg dit dat A en B óf 6 en 1 óf 3 en 2 moet wees, maar slegs $6 + 1 = 7$. Om dit te bevestig, kontrolleer dat $61 \times 16 = 976$.

7. **Antwoord 004**

   Die middellyn van die sirkels stem ooreen met die hoogte van die reghoek, wat 5 is. Die afstand tussen die sirkels se middelpunte is gelyk aan die breedte van die reghoek minus twee keer die radius (of een keer die middellyn), dit is, $9 - 5$, wat 4 gee.